

Experiment 9: Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Clustering

March 27, 2026

Name: Monesh M
Roll No: 3122235001084

1 Objective

To implement and analyze the performance of clustering algorithms (K-Means, DBSCAN, and Hierarchical Agglomerative Clustering) on the Human Activity Recognition (HAR) dataset.

2 Dataset

The dataset consists of 10,299 samples, each containing 561 features recorded from smart-phone sensors (accelerometer and gyroscope). There are 6 actual activity classes: Walking, Walking Upstairs, Walking Downstairs, Sitting, Standing, and Laying.

3 Methodology

- **Preprocessing:** Features were standardized to zero mean and unit variance.
- **Dimensionality Reduction:** PCA was used to project the 561D space into 2D for visual analysis.
- **K-Means:** The optimal number of clusters was determined using the Elbow method and Silhouette scores.
- **DBSCAN:** A density-based approach was implemented to identify core samples and noise.
- **Hierarchical Clustering:** Agglomerative clustering with Ward's linkage was used to build a dendrogram.

4 Results and Observations

4.1 K-Means Analysis

The Elbow method (Figure 1) indicated a significant reduction in inertia (WCSS) around $k = 6$, which corresponds to the known number of activities.

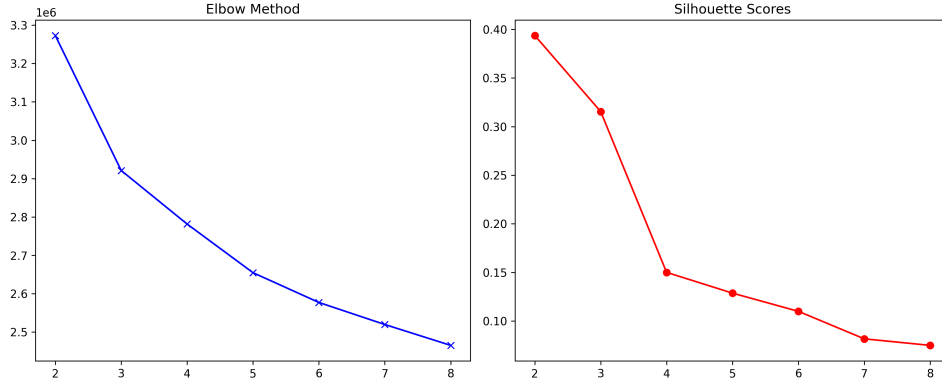


Figure 1: Elbow Method and Silhouette Scores for K-Means.

4.2 Clustering Performance Metrics

Table 1: Clustering Performance Comparison

Algorithm	Silhouette	ARI	NMI
K-Means ($k = 6$)	0.110	0.420	0.559
DBSCAN	0.119	0.080	0.154
Hierarchical	0.117	0.460	0.601

5 Visualizations

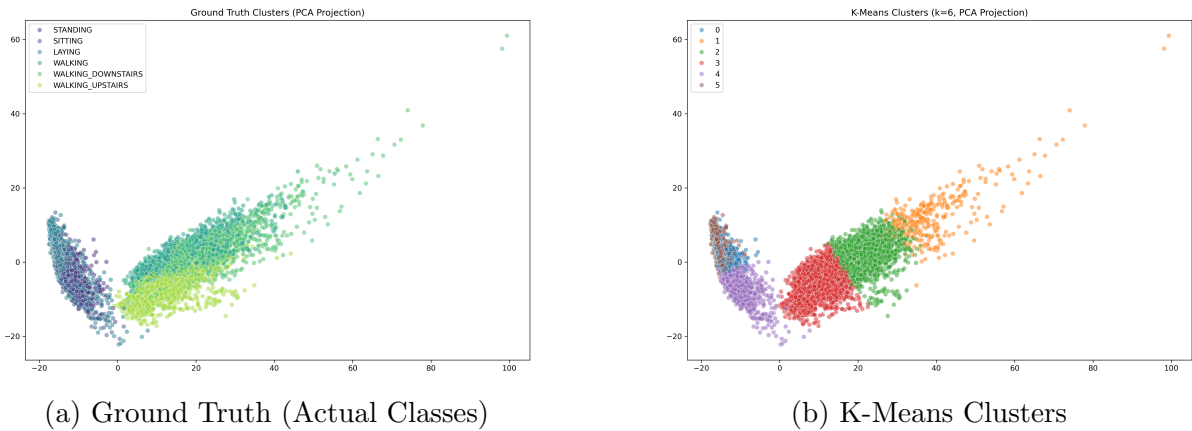


Figure 2: PCA Comparison of True Labels vs. K-Means Clusters

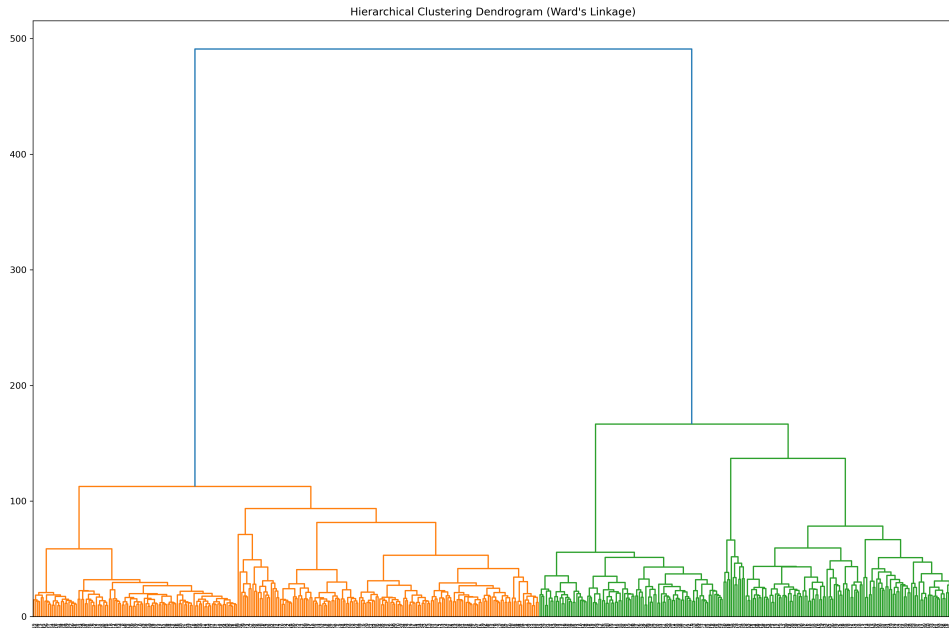


Figure 3: Hierarchical Clustering Dendrogram (Ward's Linkage)

6 Conclusion

The analysis reveals that Hierarchical Agglomerative Clustering with Ward's linkage outperformed K-Means and DBSCAN in terms of alignment with ground truth labels (ARI of 0.46 and NMI of 0.60). DBSCAN struggled due to the high-dimensional nature of the dataset.